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GENERAL PROBLEMS OF CORROSION

Atmospheric Corrosion of Metals in Regions of Cold and Extremely Cold Climate (a Review)

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Abstract—The results of atmospheric corrosion tests on a series of metals and alloys in marine and industrial atmospheres of the Earth's regions with cold and very cold climate (Antarctic, sub-Arctic, Russian Far East) are considered. The class of most dangerous corrosive damage includes special types such as pitting, exfoliation corrosion, crevice corrosion and corrosion-induced cracking. Long-term prognosis is made concerning the influence of global warming on the atmospheric corrosion in cold climate regions.

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INTRODUCTION

Despite relatively low velocities of metal corrosion in cold climate as compared to temperate, the problem of significant corrosion-related losses in regions with cold climate is not less urgent. Suffering of this damage are primarily the cultural values of the Arctic territories of Norway, Russia, Canada and United States (including Novaya Zemlya, Franz Joseph Land, Spitsbergen, Svalbard, Nunavut, Greenland, and Alaska) and structures on the Antarctic Peninsula and sub-Antarctic islands [1, 2]. Nevertheless, the number of investigations in the field of corrosion in cold climate is still insufficient, being significantly smaller than that in the regions with other types of climate. As a rule, the interest in this research is concentrated on chloride-stimulated marine corrosion, since the main human activity in sub-Arctic and sub-Antarctic regions is related primarily to the sea.

Only several corrosion testing programs involving investigations in under cold climate conditions have been completed so far, including Russian test programs on the Far East [3, 4] and in Antarctica (Mirnyi station) [5], joint Norway–Russia Arctic program [5–10], Australian program of Antarctic tests carried on several stations (including intracontinental), Russian investigations on the Vostok station, American program at the Amundsen–Scott south-pole station [11, 12], and corrosion tests on the northern stations of Norway, Sweden, and Canada according to their national and joint (Scandinavian) programs [13–15]. Corrosion tests at separate locations with cold climate were performed within the framework of ISOCORRAG program by the Technical Committee 156, which is involved in the development of international standards in atmospheric corrosion [16–18], and MICAT project [19–21].

CORROSIVITY OF ANTARCTIC ATMOSPHERE

According to the Russian State Standard (GOST 25650–83) [22], the territory of Antarctica is subdivided in macroscopic zones with very cold (VCL), cold (CL), and temperate (T) climate. Within these macroscopic zones, various climate regions and representative sites have been selected as indicated in Fig. 1. The coast of Antarctica mostly belongs to three climate regions: temperate with very strong wind (T1, representative sites: Mirnyi and Molodyozhnaya stations), temperate with moderate wind (T2, representative sites: Bellingshausen station and Adelaide Island), and cold with moderate wind (CL2, representative site: Halley station). A very small part of the Antarctic coast belongs to cold climate with strong wind (CL1, representative sites: Pionerskaya and Mizuho stations). Continental regions typically belong to very cold climate with moderate wind (VCL2, representative sites: Amundsen–Scott south-pole station and Vostok station). Finally, there are sites of very cold climate with very strong wind (VCL1, representative site: Byrd station).

Table 1 gives the values of monthly average temperature (T, °C), relative humidity (RH, %), and wind velocity (V, m/s) at various representative climate sites of Antarctica. Table 2 presents data on the duration of T–RH combinations influencing the corrosion process at these sites. In terms of ISO 9223, according to which the time of wetness (TOW) of a metal surface is determined as the period of time in which $T > 0$°C and $RH > 80\%$, the TOW is typically very short (ranging from 35 to 261 h/year) except for Bellingshausen station where this period reaches 2300 h/year. However, if the duration of wetting is evaluated with allowance for a temperature of –4°C and above, the period of severe corrosive conditions increases to 386–2360 h/year.

Table 3 presents data on the test sites (geographic coordinates, altitude), conditions (distance from sea, starting year, duration), and results (corrosion rates) for carbon steel (Russian program, Mirnyi station) and mild steel (Australian program of tests at various sites of Antarctica) [2, 5, 12]. According to the aforementioned classification (GOST 25650–83), the sites involved in these tests belong to regions of the following types:

T1: Mirnyi, Mawson, Robertskollen, LGB 00;

T2: Rothera (Adelaide Island, Antarctic Peninsula), Signy Island;

CL1: LGB 10;

CL2: Wanda, Scott's Hat (cape Evans);

VCL1: South Pole, Vostok, LGB 35.

Since the corrosion tests in Antarctica are very seldom and extremely difficult to realize, any new information concerning such investigations is of considerable interest. The Australian program of corrosion testing was carried out both in the continental regions and on sub-Antarctic islands (Macquarie, Hurd, Marion, and Signy islands). The islands are characterized by a sub-Antarctic climate, which differs very strongly from that in the Antarctic continental regions. On these islands, the climate is generally cold (with summer temperatures from +10 to slightly below 0°C). Fast ice is rarely formed. Winds are typically strong and gusty. Being located far from industrial territories, the islands are characterized by clean atmosphere with natural background contamination level. Only the Hurd Island has an active volcano, which can eject sulfur dioxide into atmosphere.

The most important factor that determines the climate of islands is their location relative to the so-called polar front (previously also known as the Antarctic convergence). The polar front is a temperature boundary between cold water surrounding Antarctic and moderately cold water in the northern direction. Islands situated on the south from this boundary are characterized by a generally lower temperature of both water and air. The Marion Island (~18 × 15 km) is situated to the north from the polar front, the Hurd Island (~54 × 25 km) is to the south from this front, and the Macquarie Island (~34 × 6 km in the widest part) is about 1500 km from Tasmania.

The tests were performed on carbon steel (composition, wt%: C, 0.15; Mn, 0.42; Si, trace; P, 0.024; S, 0.013) and mild steel (composition, wt%: C, 0.18; Mn, 0.76; Si, 0.05; P, 0.017; S, 0.018; Mo, 0.003; Sn, 0.003; Al, 0.011; Cr, 0.125; Ni. 0.295; Cu, 0.235). The rate of corrosion for these steels on the Antarctic continent varied from 27.1 μm/year on the coast of Rothera point to 0.05 μm/year at the Vostok station and 0.03 μm/year at the South Pole. Note that steel samples were continuously exposed at the South Pole for about 3.7 years and it was unknown whether the rate of corrosion in the first year was greater than the average (0.3 μm/year). The rate of corrosion on the Signy Island (36.4 μm/year) was even greater than that on the Rothera point.

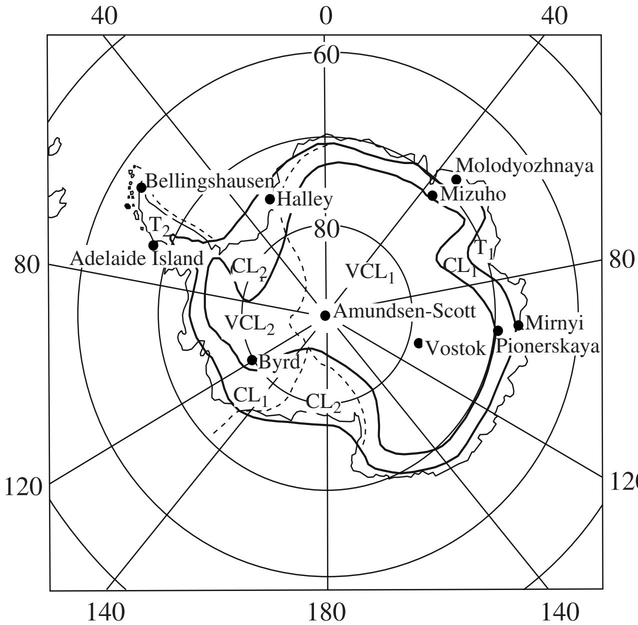

Fig. 1. Subdivision of Antarctica into climate zones (solid curves) and local climate regions (dashed curves) with respect to the parameters influencing the corrosion of materials and related articles (according to GOST [22]): (VCL1) very cold climate with strong and very strong wind; (VCL2) very cold climate with weak and moderate wind; (CL1) cold climate with strong wind; (CL2) cold climate with moderate wind; (T1) temperate climate with very strong wind; (T2) temperate climate with moderate wind and high humidity. Black circles indicate representative sites.

The atmospheric corrosion tests at the Mirnyi station were initiated by the Institute of Physical Chemistry (Frumkin Institute of Physical Chemistry and Electrochemistry since 2005) jointly with the Institute of Arctica and Antarctica [5]. The experiment was started on May 9, 1989 (during operation of the 34th USSR Antarctic Expedition) and terminated in April 16, 1990, so that the duration of exposure was about one year (342 days). Table 4 gives the monthly average climate parameter for this period of time.

The initial period of test was characterized by low air temperatures (T_{max} = –7°C). The deviation from normal average temperature in May 1989 was –2.9°C. However, the second half of 1989 was characterized by monthly average temperatures well above the many-year average data: August, +2.9°C; September, +5.9°C; October, +4.5°C. No such an increase in the temperature was observed before for the entire 34-year period of existence of the Mirnyi station. Shortly before the onset of corrosion tests, the Davis Sea exhibited complete freezing. The ice sheet began to break on November 20, 1989, at a distance of about 22 km away from the Mirnyi station. By December 4, the fraction of immobile ice on the visible area decreased (in conventional terms) to rate 7. From this moment on, the probability for sea-water aerosols to reach the test site began to increase. In addition to an east wind (which predom-

Table 1. Monthly average temperature (T, °C), relative humidity (RH, %), and wind velocity (V, m/s) at some representative sites of Antarctic coast [5] and Scott Base station [11] (climate region types according to GOST 25650-83)

Station (climate region type)	Parameter	Month											
		1	2	3	4	5	6	7	8	9	10	11	12
Mirnyi (T1)	T, °C	–1.6	–5.3	–10.1	–13.9	–15.3	–15.4	–17.7	–17.5	–17.4	–13.5	–7.1	–2.4
	RH, %	71	68	68	70	73	74	74	73	72	68	68	71
	V, m/s	7.7	9.2	11.1	12.6	13.3	13.3	12.9	13.0	12.2	10.5	9.9	8.6
Molodyozhnaya (T1)	T, °C	–0.4	–4.1	–8.4	–11.5	–14.0	–16.7	–18.3	–18.8	–17.7	–13.6	–6.7	–1.3
	RH, %	62	61	67	69	66	65	66	70	69	66	63	61
	V, m/s	5.3	8.5	12.7	14.4	15.5	13.5	11.4	10.6	9.9	9.4	8.2	5.9
Adelaide Island (T2)	T, °C	0.6	0.0	–1.6	–4.6	–6.5	–8.8	–9.7	–11.3	–8.0	–5.3	–2.3	0.1
	RH, %	79	78	80	80	80	80	80	81	81	81	77	78
	V, m/s	3.6	4.4	6.6	6.5	6.5	7.2	7.7	7.4	9.2	8.6	6.5	4.4
Bellingshausen (T2)	T, °C	1.1	1.2	0.1	–1.6	–4.3	–6.3	–7.0	–7.4	–4.4	–2.4	–1.1	0.3
	RH, %	84	87	87	87	86	86	88	88	89	88	87	87
	V, m/s	6.2	7.0	7.0	7.8	7.1	7.5	7.1	7.8	7.9	7.9	6.8	6.6
Halley (CL2)	T, °C	–4.8	–10.0	–16.4	–20.1	–23.9	–26.5	–29.2	–28.9	–26.1	–19.2	–11.8	–5.4
	RH, %	84	83	81	79	78	77	76	76	77	81	82	83
	V, m/s	5.1	5.8	6.9	7.0	6.4	7.0	6.9	6.9	7.6	7.6	6.7	5.5
Scott Base (CL2)	T, °C	–5.0.	–10.8	–20.4	–24.2	–27.4	–26.2	–29.4	–31.0	–28.1	–22.7	–12.0	–5.4

inated in December 1989 and January 1990), there were periods of continental winds (typically, strong) in the south direction. Table 5 presents the results of corrosion tests at the Mirnyi station.

As can be seen from the data presented above, carbon steel is subject to corrosion even at the Vostok and South Pole stations of continental Antarctica, where positive (centigrade) temperatures are not reached over the entire year. At the coastal stations, the rates of corrosion reach 4.4–10.8 μm/year (Scott's Hat) and 7.7 μm/year (Mirnyi), which is obviously inconsistent with an extremely small TOW of metal surfaces at the Antarctic coast, if this is evaluated in terms of ISO 9223 (i.e., at temperatures above 0°C).

Table 2. Duration of T–RH combinations (h/year) and time of wetness (at $RH > 80\%$) at some representative sites of Antarctica (climate region types according to GOST 25650-83)

Station (climate region type)	T, °C	RH, %		TOW, h
		81–90	91–100	
Mirnyi (T1)	– 4–0	257.8	283.4	541.2
	>0	52.4	22.9	75.3
Molodyozhnaya (T1)	– 4–0	236.4	150.0	386.4
	>0	31.2	4.2	35.4
Halley (CL2)	– 5–0	362.7	426.6	789.3
	>0	20.1	19.3	39.4
Bellingshausen (T2)	– 4–0	1016.9	1821.9	2838.8
	>0	629.8	1669.9	2299.7
Adelaide Island (T2)	– 4–0	1079.8	1280.0	2359.8
	>0	119.4	142.3	261.7

The corrosivity of atmosphere in non-industrial coastal regions is determined not only by the TOW, but it also depends on the concentration of marine aerosols—in particular, chlorides deposited from air onto the metal surface [23–27]. The rate of steel corrosion at the Mirnyi station is comparable with (and that of copper even exceeds) that observed on the Aion Island and Apapel'khino Peninsula (Far East), where the Cl^- concentrations amounted to 33.8 ad 20.1 mg/(m^2 day) and the TOW exceeds 1000 h/year [4, 26] (the Far-East data were obtained using the dry-fabric method and recalculated to the wet-candle values [25, 28]). However, if the TOW would be determined for $T > -4$°C, the corresponding value for the tests at Mirnyi would increase by a factor of more than 7 to reach 987 h/year (in the test period) [5]. This value is in better agreement with the rates of corrosion observed in these tests.

The main factors determining the level of corrosivity of an open atmosphere are the TOW, air temperature, relative humidity and the concentration of contaminants (sulfides in industrial regions and chlorides in coastal regions). The effect of the first three factors can be taken into account using a generalized characteristic called the temperature–humidity complex (THC), which is defined as the duration of exposure to a certain combination of air humidity and temperature. Data on the THC and related corrosion rates in the regions of corrosion testing arte of considerable practical interest [25–27]. Based on an analysis of the results of many-

Table 3. Data on the test sites, conditions, and average corrosion rates in various regions of Antarctica [2, 5, 12]

Test site	Geographic coordinates		Elevation, m	Distance from sea, km	Test start, year	Test period, days	Corrosion rate, μm/year
	Latitude	Longitude					
Wanda station	–77.52	161.57	130	44	1983	360	0.87
Scott's Hat (cape Evans)	–77.63	166.4	5	0.2	1983	365	10.83
"	–77.63	166.4	5	0.2	1983	365	10.20
"	–77.63	166.4	5	0.2	1983	1053	4.42
Mirnyi station	–66.55	93.02	39.9	0.25	1989	342	7.7
Macquarie Island	54.50	–	7.5	<1	1994	721	222
Hurd Island	53.10	–	<10	<1	1994	341	55.30
Marion Island	46.86	–	<10	<1	1994	376	31.90
Signy Island	–60.717	–45.60	<10	<1	1994	365	36.40
Rothera Point	–67.567	~–70	<10	<1	1994	365	27.10
Mawson station	–67.60	62.883	<10	<1	1994	372	3.35
Robertskollen (SANAE)	–68.483	~0	400	120	1994	377	0.70
Site LGB 00	–68.65	~61	1830	186	1994	789	0.18
Site LGB 10	–71.283	~60	2616	390	1994	777	0.10
Site LGB 35	–76.05	~58	2340	780	1994	380	0.13
Vostok station	–78.45	106.85	3488	1200	1994	347	0.05
South Pole	–90	–	2800	1300	1994	1364	0.03

Table 4. Data on the temperature (monthly average, minimum, maximum), relative humidity (*RH*), wind (velocity *V* and bearing direction), and time of wetness (TOW for $T > 0°C$, $RH > 80\%$) at the Mirnyi station during corrosion tests (May 1989–April 1090) on various metals

Month, year	Air temperature, °C			Relative humidity, %			Wind		TOW, h/year
	Average	Minimum	Maximum	Average	Minimum	Maximum	*V*, m/s	Bearing	
05.1989	–18.0	–29.1	–7.0	76	52	98	11.8	E	0
06.1989	–16.7	–30.4	–7.8	76	42	100	14.1	E	0
07.1989	–16.9	–30.7	–5.4	72	48	98	13.2	SSE	0
08.1989	–14.2	–29.3	–0.7	67	36	97	14.6	ESE	0
09.1989	–11.3	–20.3	–1.4	66	35	98	14.3	E	0
10.1989	–9.0	–20.4	–2.3	70	38	99	13.7	E	0
11.1989	–10.5	–22.5	–1.7	60	36	99	8.0	E	0
12.1989	–1.2	–8.5	3.6	81	41	100	11.3	E	51
01.1990	–1.6	–9.7	2.7	73	42	99	9.5	E	42
02.1990	–6.2	–14.4	1.5	74	42	98	8.0	SSE	0
03.1990	–11.1	–24.8	–1.5	76	51	97	11.0	SSE	0
04.1990*	–12.9	–20.4	–5.9	73	53	94	14.0	SE	0

* Over a period of April 1–16, 1990.

year measurements of the temperature and relative humidity of air at 3053 weather stations on the territory of the former USSR, we have previously systematized THC data for various climate regions ranging from very cold to very hot (according to GOST [29]) and for the coastal regions of Arctic Ocean, Pacific Ocean, and Okhotsk Sea [26].

In continental regions, the THC values are rather large, ranging from 200 to 1500 h/year. There is a tendency to increase in THC on the passage from cold to temperate wet and warm climate, followed by a decrease for a dry hot climate. In the regions of cold climate, the maximum THC values are observed within 0–15°C. In coastal regions, THC increases by about 1000 h/year as

Table 5. Corrosion rates of carbon steel (St3), copper, cadmium, and an aluminum-based alloy (D16) and the content of chlorine ions ([Cl⁻]) in water-soluble corrosion products for one-year corrosion tests at the Mirnyi station

Material	Corrosion rate		$[Cl^-]$, mg/m^3
	g/(m^2 year)	μm/year	
Steel St3	60.8	7.8	536
Copper	9.9	1.1	175
Cadmium	2.7	0.3	175
D16 alloy	2.2–4.8	0.8–1.77	143

Table 6. Total corrosion rates [g/(m^2 year)] of metals in continental regions with different climate and conditionally clean air and in coastal regions [26]

Region, climate	Steel	Copper	Aluminum
Continental zone			
Very cold	5–20	0.5–2.0	–
Cold	10–30	0.5–2.5	–
Arctic east	10–50	0.8–7.0	–
Arctic west	15–50	2–7	–
Temperate cold	15–50	0.8–7.0	–
Temperate wet	40–70	2.5–7.0	0.1–0.2
Coastal zone			
Arctic Ocean	20–300	3–20	0.6–5.5
Bering Sea	100–250	8–20	0.7–2.3
Okhotsk Sea	30–250	4–25	0.4–2.6

compared to the continental regions and changes within an order of magnitude (400–5000 h/year). The maximum THC values at the Pacific Ocean coast are observed at temperatures within 0–10°C.

We have analyzed the interval of THC values for continental regions with conditionally clean atmosphere jointly with experimental data on the corrosion of carbon steel, copper, and aluminum, by analogy with

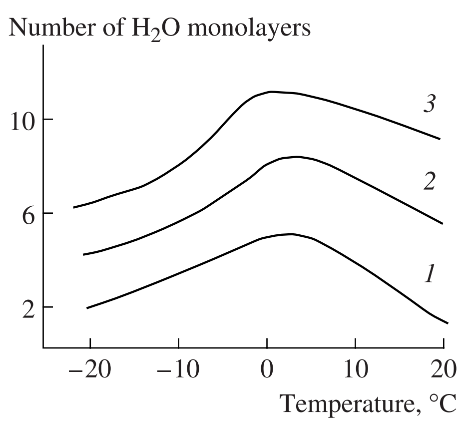

Fig. 2. Plots of the amount of physically adsorbed water on silver versus temperature at RH = 40 (*1*), 60 (*2*), and 80% (*3*).

the joint analysis performed previously for the seashore regions (Table 6) [6]. The large spread of data on the corrosion of metals in coastal regions is related to considerable variations in the level of atmospheric chlorides, which significantly accelerate corrosion in wet air. This level depends on the salinity of water, the velocity, duration and direction of sea winds (transporting salt aerosols), distance to the sea, ground relief, etc. Such a variety hinders establishing a unique relationship between THC and the corrosion of metals. For seashore territories, rather approximate intervals of corrosion rates are obtained (Table 6) even using the reliable THC data and the results of natural corrosion tests.

The atomic and molecular species formed by adsorbed water on the surface of solids are subject to discussion for a long time [30]. In particular, the temperature interval of the phase transitions of adsorbed water into various modifications of ice is still not established.

Laboratory investigations of the initial corrosion stages for cadmium and silver in wet air at negative and positive temperatures showed that the time necessary for attaining the adsorption equilibrium strongly depends on the temperature and is maximum near the point of the phase transition in water. The appearance of a maximum in the adsorption of water vapor on metals, which is observed in the region from –5 to +5°C at any relative humidity, is related not as much to the freezing of waster as to the fact that the amount of adsorbed water is a decreasing function of the temperature and the increasing function of vapor pressure (Fig. 2) [31],

In order to more strictly describe the state of a metal surface in humid air at low temperatures, is expedient to construct the adsorption isobars (amount of adsorbed water versus temperature at a preset vapor pressure) [32, 33]. An analysis of the adsorption isobars for water on silver shows that the amount of adsorbate at any fixed humidity rapidly decreases with increasing temperature. The adsorption of water on silver was also characterized by isosteres (temperature dependences of the equilibrium vapor pressure at a constant amount of adsorbed water). The isosteres exhibit no clearly pronounced bending points in the region of phase transition temperatures (from –18 to +14°C) corresponding to a change in the number of adsorbed water monolayers from 5 to 10, which is an indirect evidence for the absence of water freezing in the adsorbed phase at low temperatures. This circumstance implies the possibility of corrosion development at negative temperatures. This conclusion was confirmed by the results of natural tests using samples of deposited metal films on piezoquartz resonator plates.

We have also studied the effect of daily variations in the relative humidity and temperature on the adsorption of water on the surface of aluminum and magnesium and the kinetics of rural atmospheric corrosion in regions with temperate climate [33, 34]. The metal samples were exposed in a box with baffle walls (modeling ventilated unheated room). In summer (July to August), the air temperature varied within 8–25°C, while in winter (December to January) it was always negative (from –2 to –26°C). In autumn (September to November), the temperature could be both positive and negative (from –6 top +16°C). Despite the rather large interval of temperature variations (from –20 to +26°C), the corrosion of both aluminum and magnesium under adsorbed water layers proceeded at positive temperature almost at the same rate as at negative temperatures [34].

The results of earlier investigations also showed that the adsorbate did not exhibit freezing in the range of mono- and polymolecular adsorption. The phase transition took place only in a capillary-condensation region of the adsorption isotherm. This behavior is conventionally interpreted in terms of the theory of capillary condensation, according to which a decrease in the temperature of solidification of a liquid is related to a decrease in the saturated vapor pressure over meniscus in the capillary [35].

It was suggested that a decrease in the freezing (melting) point of an adsorbate can be interpreted proceeding entirely from properties of a substance in this state, where the conditions of crystallization (melting) are different from those in the bulk liquid. Based on such theoretical calculations, Fletcher [36] obtained the temperature dependence of the thickness of a quasi-liquid water film on the surface of ice. At –20°C, the thickness of such a film is still on the order of several nanometers.

CORROSIVITY OF SUB-ARCTIC ATMOSPHERE

The aforementioned corrosion tests in Antarctica were carried out in atmospheres with low and very low levels of contamination. In contrast, the northern territories of Norway and Russia, which traditionally featured extensive development of mining and metallurgical industries, are presently characterized by very strong negative effect on both the environment and structural materials. In Norway, the ecological damage was most pronounced near the city of Rogos, where copper was produced since 1646. In 1977, this city built entirely of wood was classified by UNESCO into the Book on World Heritage Sites. Then, the copper mines were closed, after which the site of maximum sulfur dioxide emission into the atmosphere in Norway for the next decade (until 1987) was at Sulitjelma (67°08'N) explored since 1886.

The corrosion tests on steel, zinc, and copper were performed at three sites (Sandnes, Furulund, Lomi) [37, 38]. The parameters of climate at these sites were virtually identical and the monthly average levels of SO_2 could significantly vary: from 600 μg/m^3 in winter to 6 μg/m^3 in summer (in this period, the foundries were not operated). After termination of the copper production, the level of SO_2 in this region stabilized on a level of ~2 μg/m^3.

The program of atmospheric corrosion tests included one-month exposures of carbon steel, and one-year exposures of steel, zinc, and copper. A comparison of the results of one-year exposures in 1984–1986 (high atmospheric SO_2 level) to the data for 19087 (low SO_2 content after shut-up of the foundries) clearly demonstrates a decisive effect of SO_2 on the corrosion of metals (Table 7). Indeed, the rate of corrosion decreased to 60 g/(m^2 year) for steel and 3 g/(m^2 year for zinc. These results well agree with the other data reported for three sites with close air temperatures. The effect for copper was not as pronounced, since an increase in the rate of copper oxidation in the first year of testing was related mostly to an increase in the concentration of ozone in the clean air [39–41]. Evidently, exposures on longer terms are necessary in order to reveal the accelerating effect of SO_2 on the corrosion of copper.

Table 7. Annual average corrosion rates [g/(m^2 year)] of carbon steel, zinc and copper tested at three sites near Sulitjelma

Material	Period (month, year)	Test site		
		Sandnes	Furulund	Lomi
Carbon steel	01.1984–01.1985	592	493	–
	01.1985–01.1986	449	431	–
	03.1986–03.1987	549	511	488
	03.1987–03.1988	67	60	66
Zinc	01.1984–01.1985	60	60	–
	01.1985–01.1986	46	50	–
	03.1986–03.1987	50	43	33
	03.1987–03.1988	3	3	4
Copper	01.1984–01.1985	9.0	6.8	–
	01.1985–01.1986	10.4	9.4	–
	03.1986–03.1987	7.3	6.5	5.8
	03.1987–03.1988	5.5	5.8	5.5

The corrosion behavior of steel and zinc can be described by simple linear regression equations that relate the annual metal loss (ML, g/(m^2 year)) versus effective SO_2 concentration [μg/m^3] in air:

$$ML_{\mathrm{steel}} = 1.2[SO_2] + 63, \quad (1)$$

$$ML_{\mathrm{zinc}} = 0.12[SO_2] + 4.1. \quad (2)$$

Note that both equations are valid only in the region where the tests have been carried out, since the climate parameters (which are the same at the three test sites) determine the coefficients and free terms in these relations. The effect of climate was evaluated using the results of one-month tests in 1986. A comparison of the monthly average SO_2 concentration to the rate of steel corrosion over the same period of time showed that the corrosivity of SO_2 decreased with decreasing temperature, especially in November–December 1986. The maximum retardation of corrosion was observed in December, where the monthly average temperature was as low as –10.2°C.

Table 8. Linear regression equations describing the metal loss (ML, g/(m^2 month)) caused by atmospheric corrosion of carbon steel at Sulitjelma depending on the effective SO_2 concentration ([SO_2], μg/m^3) for various temperatures of adsorbed water freezing (T_F) on the metal surface

Effective [SO_2] determined	Regression equation	R^2
With neglect of freezing	$ML = 0.19[SO_2] + 9.1$	0.74
For $T_F = 0°C$	$ML = 0.27[SO_2] + 20.8$	0.84
For $T_F = -2°C$	$ML = 0.27[SO_2] + 13.2$	0.94
For $T_F = -4°C$	$ML = 0.24[SO_2] + 13.1$	0.92
For $T_F = -6°C$	$ML = 0.27[SO_2] + 13.1$	0.89

In order to take into account the temperature effect, an additional parameter (C_{eff}) was introduced so as to characterize the effective concentration of sulfur dioxide. The daily SO_2 concentration C_d was multiplied by the fraction of time (t, h) corresponding to a positive or slightly negative temperatures, at which the adsorbed water film on the metal surface is not frozen: $C_{eff} = C_c(t/24)$. The proposed threshold temperatures (0, –2, –4, and –6°C) were evaluated by means of the regression analysis of the experimental values of metal losses as functions of the effective SO_2 concentration (Table 8). Judging from the high correlation coefficients ($R^2 = 0.92 - 0.94$), the anticipated threshold temperature falls in the interval between –2 and –4°C.

Presently, the most intense sources of SO_2 evolution on a territory at the Russia–Norway border are the Russian nickel producing enterprises at Nikel and Zapolyarnyi. This activity was started in the 1930s and produced a strong detrimental effect on the nature and state of ecology on both sides of the border [42–46].

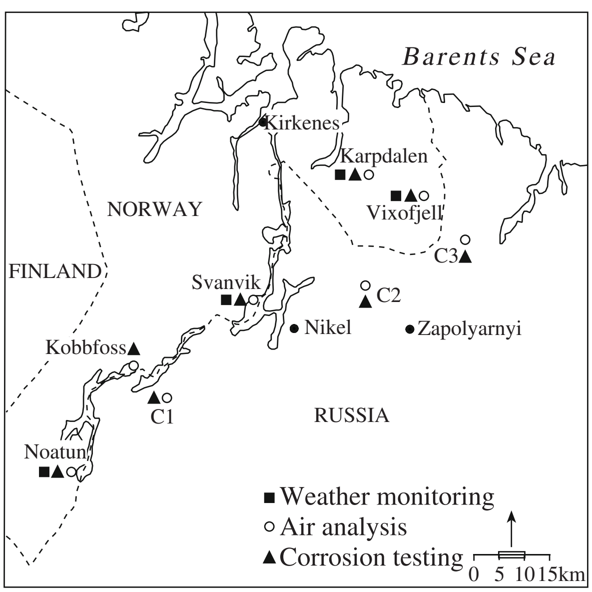

Fig. 3. The map of a territory at the Russia–Norway border indicating the sites of atmospheric corrosion tests and the measurements of weather parameters and sulfur dioxide concentration in air.

Corrosion tests at Sulitjelma were previously carried out in 1984–1988 [17]. Taking into account the results of those tests, two stages of more detailed investigations into atmospheric corrosion have been organized jointly by the Norwegian Institute for Air Research (NILU) and the Institute of Physical Chemistry (Now the Frumkin Institute of Physical Chemistry and Electrochemistry) at various sites in the region at the Russia–Norway border in 1990–1991 [6, 7] and 1992–1994 [8–10, 38]. Since this region (Fig. 3) was significantly greater than that studied previously at Sulitjelma, the climate parameters varied rather significantly and the influence of marine aerosols was significant at the northern sites.

Table 9 presents data on the annual corrosion rates of carbon steel and zinc tested in 1990–1991 at various sites near the Russia–Norway border. In comparison to the rates of corrosion [60 g/(m^2 year) for steel and 3 g/(m^2 year) for zinc] observed in Sulitjelma after shut-up of the foundries, the values obtained at all test sites along the Russia–Norway border are higher. The maximum corrosion rates have been observed at Vixofjell and C2, where they are greater than the corresponding values at Sulitjelma by a factor of 5.8 and 5.1 for steel and by a factor of 8 and 7 for zinc, respectively. The annual average SO_2 levels at these test sites amount to 37.1 and 56.9 μg/m^3, respectively.

Figure 4 shows a plot of the corrosion rate of carbon steel versus SO_2 concentration in air in 1990–1991 for various test sites on the Russian and Norway sides of the border. As can be seen, the presence of chlorides is clearly manifested by an additional contribution to the rate of corrosion at Vixofjell. The presence of chlorides was confirmed by means of an analysis of the absorption filters exposed monthly in special collectors of aerosol species, which had been developed at the NILU. The amount of chlorides collected at Vixofjell was approximately four times the average value for Svanvik. The results of two-year tests (July 1992–July 1994) at two sites in Norway and three sites in Russia (Table 10) showed that the rates of corrosion for both steel and zinc decrease with the time, but still remain rather high at the sites with increased concentrations of SO_2 (Vixofjell, C2, and C3).

The results of one-month exposures confirmed that metals tested at the Russia–Norway border, like those tested previously at Sulitjelma, are subject to corrosion even in winter, that is, when the temperature is negative throughout the test period. The results of these experiments for steel were processed generally in the same way as were the data obtained in Sulitjelma, but the temperature effect was evaluated by selecting various threshold temperatures for the subsequent TOW calculations. It was found that the maximum correlation coefficient was obtained for the TOW values calculated relative

Table 9. Corrosion rates [g/(m^2 year)] of carbon steel and zinc tested at various sites near the Russia–Norway border

Period (years)	Material	Vixofjell	Karpdalen	Svanvik	Noatun	C1	C2	C3
1990–1991	Steel	308	180	108	78	93	261	214
1992–1993	″	347	–	145	–	163	306	264
1990–1991	Zinc	24	12	9.6	5.4	6.4	19	15
1992–1993	″	18	–	7.1	–	8.3	21	15

Note: Here and in Table 10: (C1) Nikel (SW, 25 km); (C2) Nikel (NE, 15 km); (C3) Pechenga.

Table 10. Corrosion mass losses [g/m^2] of carbon steel and zinc tested for one (July 1992–July 1993) and two years (July 1992–July 2004) at various sites near the Russia–Norway border

Material	Vixofjell		Svanvik		C1		C2		C3	
	1 year	2 years	1 year	2 years	1 year	2 years	1 year	2 years	1 year	2 years
Steel	347	539	145	206	163	220	306	466	264	392
Zinc	18.0	26.8	7.1	9.5	8.3	11.2	21.3	33.4	15.1	27.1

to a lower threshold temperature of –4°C (Table 11). Thus, the results of various investigations suggest approximately the same temperature of freezing for electrolyte on the metal surface. It should be noted that the free term A in regression equations presented in Table 11 h is negative and, hence, these equations can be used for calculations only in that particular interval of parameters of the environment, for which they are derived.

Upon termination of the second stage of atmospheric corrosion tests, the annual data were used to construct the dose–response functions and to plot the maps of corrosion rates for carbon steel and zinc for various scenarios of variation of the sulfur dioxide level [10, 38]. Corrosion situations in the region under consideration were modeled in terms of equation of the type $ML = A + B$ TOW [SO_2], which took into account the possibility of corrosion development at temperatures above –4°C.

CORROSIVITY OF ATMOSPHERE IN FAR EAST TERRITORIES

When the Institute of Physical Chemistry (Now the Frumkin Institute of Physical Chemistry and Electrochemistry) initiated atmospheric corrosion tests in 1980 in the eastern territories of Russia, almost no data were available on the corrosion behavior of metals in the climate of East Siberia and Far East. Investigation of the corrosion behavior of metals and alloys under such conditions was both of basic significance and of considerable practical importance in view of the intense exploration of those territories, development of gas-and-oil industries, construction of the Baikal–Amur railroad, and large industrial complexes.

Figure 5 shows a map of the Far East with the sites of atmospheric corrosion testing stations. Tables 12 and 13 present data on the positions of these stations, environmental conditions, and corrosion rates of carbon steel, zinc, copper, and aluminum [3, 4]. For the comparison, Tale 13 also shows data for the sub-Arctic station (Northern Corrosion Station, NCS) on the Barents Sea coast at Murmansk (Dal'nie Zelentsy). The standard characteristics of climate were calculated using the results of many-year weather monitoring. The samples were exposed on special holders, which were oriented toward the north and sloped at an angle of 45° to the horizontal plane.

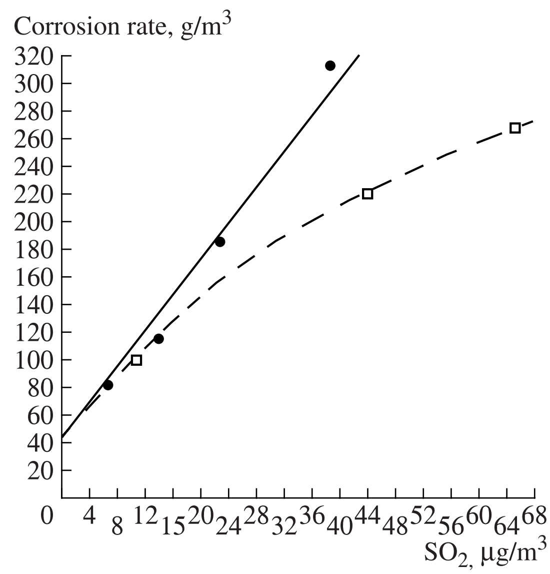

Fig. 4. A plot of the corrosion rate of carbon steel versus SO_2 concentration in air for various test sites on the (□) Russian and (●) Norway sides of the border.

Table 11. Coefficients in the regression equation $ML = (A + B[SO_2] + C[Cl^-])TOW$ (R^2 corresponds to TOW values calculated in temperature intervals with different lower threshold temperatures)

TOW threshold temperature	A	B	C	R^2
$T > -2°C$	–0.26	0.0088	0.0001	0.79
$T > -4°C$	–0.16	0.0077	0.00004	0.96
$T > -6°C$	–0.07	0.0067	0.00001	0.81

Note: Units of measure: *ML*, g/(m^2 year); [SO_2], μg/m^3; [Cl^-], μg/(m^3 day); *TOW*, h/month

A specific feature of the Far East climate is a broad range of the annual average temperature and relative humidity. Indeed, the annual average temperature changes from –19.4°C in the central region of Yakutia to +5.6°C in Primorskii Krai, and the *RH* varies from 62 in some regions of Khabarovsk Krai to 92% on the north of Yakutia. The level of atmospheric precipitation is determined primarily by the cyclone activity, which is most intense on the east and south-east coast of the Far East seas and decreases in the north-west direction, with the corresponding change in the volume of precipitation. The annual precipitation level varies from 130 mm (or even below) in the central Far East regions to 1000 mm on the east and south-east coast. The maximum precipitation is observed on Kuril Islands (more than 1000 mm) and on Kamchatka Peninsula (1400–1600 mm on the south-east coast and up to 2500 mm in the regions of geothermal activity). From the total precipitation volume, warm period accounts for 50–60% in the northern and central parts of this territory and for 80–90% in the south-east part. The time of precipitation varies from 300 to 1200 h/year.

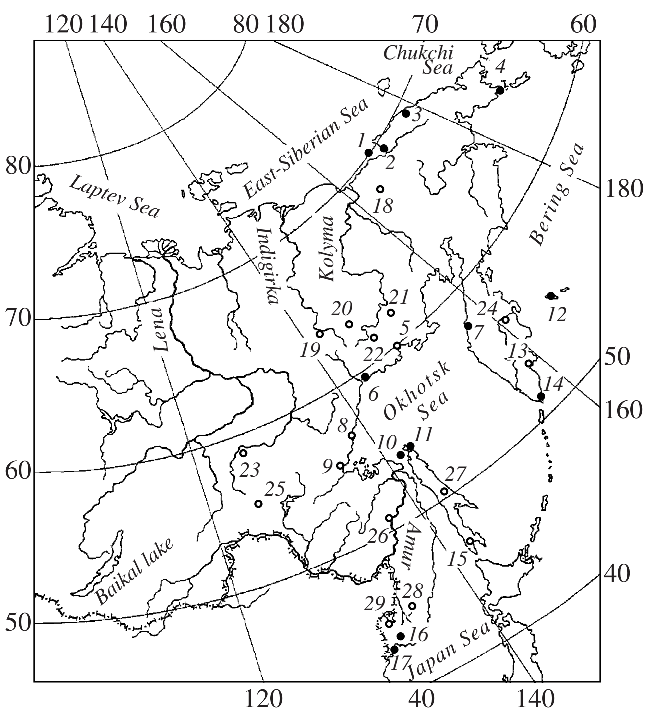

Fig. 5. A map of the Far East with the sites of atmospheric corrosion testing stations.

Coastal stations: (*1*) Aion Island; (*2*) Apapel'khino Peninsula; (*3*) Cape Shmidt; (*4*) Cape Chaplin; (*5*) Arman' village; (*6*) Okhotsk city; (*7*) Ust'-Khar'yuzovo village; (*8*) Ayan village; (*9*) Chumikhan village; (*10*) Baidukov Island; (*11*) Okha city; (*12*) Nikol'skoe village; (*13*) Petropavlovsk-Kamchatskii city; (*14*) Cape Lopatka; (*15*) Nevel'sk city; (*16*) Vladivostok city; (*17*) Cape Gamov.

Continental stations: (*18*) Bilibino city; (*19*) Oimyakon village; (*20*) Susuman city; (*21*) Atka village; (*22*) Ust'-Omchug; (*23*) Aldan city; (*24*) Klyuchi village; (*25*) Tynda city; (*26*) Komsomol'sk-on-Amur city; (*27*) Pobedino village; (*28*) Yakovlevka village; (*29*) Pogranichnyi village.

The climate characteristics of corrosion testing sites cover the entire range of their variation in the Far East. Figure 6 shows the annual variation of the air temperature in the coastal and continental regions. The maximum temperature spread for the coastal stations is about 26°C in February and about 18°C in August (the warmest month). For the costal stations, the temperature spread reaches 34°C in December and is only 13°C in July. The amplitude of temperature oscillations over one year for the coastal stations is maximum on the coast of the Arctic Ocean (Fig. 6a, points *1–3*) and minimum on islands of the Pacific Ocean (Fig. 6a, points *12–14*).

Figure 7 shows profiles of the annual variation of the relative humidity constructed using the data of many-year monitoring. In the northern regions of Far East and in some regions of Kamchatka and Sakhalin, the relative humidity during the entire year falls within 80–95% (Fig. 5, points *1–4*, *7*, *10–12*; Fig. 7a, data for Aion Island, Apapel'khino Peninsula, Cape Shmidt, Cape Chaplin, Ust'-Khar'yuzovo village, Baidukov Island, Okha city, Nikol'skoe village).

The other coastal corrosion stations are characterized by a dry winter with *RH* = 50–70% (Figs. 5 and 6a, data for Arman' village, Okhotsk city, Ayan village, Petropavlovsk-Kamchatskii city, Nevel'sk city, Vladivostok city, Cape Gamov) and wet summer wit *RH* > 80%. The interval of *RH* variation during a year for all continental stations is on the order of 15–20% (Fig. 6b), although the maximum *RH* for some stations is observed in summer (Fig. 7b, *26–29*, Komsomol'sk-on-Amur, Pobedino village, Yakovlevka village, Pogranichnyi village) and for the other in winter.

The rate of chloride deposition at the coastal stations was evaluated using the dry-fabric method and then recalculated to the wet-candle values presented in Table 13. During the corrosion tests, the atmosphere was very clean at most sites and could be classified as rural. The level of sulfur dioxide was relatively low even in big cities (see Tables 12 and 13). For this reason, the results obtained in regions with very cold climate (with the

Table 12. Test sites (geographic coordinates), environmental parameters, and rates of corrosion of structural metals in continental regions of Russian Far East with cold and very cold climate

Test site	Geographic coordinates		T, °C	RH, %	TOW, h/year	Precipitation, mm/year	$[SO_2]$, μg/m^3	Corrosion rate			
	Latitude	Longitude						Steel*	Zinc*	Copper*	Aluminum**
Oimyakon	63.47	142.82	–16.6	72	1086	215	~1	1.04	0.25	0.08	0.189
″	63.47	142.82	–16.6	72	813	215	~1	0.93	0.17	0.08	0.189
″	63.47	142.82	–16.6	72	813	215	~1	0.22	0.32	0.07	–
″	63.47	142.82	–16.6	72	273	215	~1	0.85	0.13	0.08	0.189
Susuman	62.77	148.13	–13.6	71	985	255	5	2.18	0.39	0.15	0.475
Atka	60.83	151.75	–12.0	72	987	298	~1	1.90	0.24	0.11	0.165
Bilibino	68.03	166.25	–11.0	73	692	173	~1	0.70	0.23	0.09	0.178
Ust'-Omchug	61.15	149.63	–11.0	68	1097	282	5	1.59	0.41	0.10	0.243
Aldan	58.60	125.42	–6.2	72	1333	546	5	3.20	0.77	0.20	0.205
Tynda	55.12	124.73	–6.5	72	1321	525	5	2.72	0.74	0.18	0.281
Klyuchi	–	–	–1.2	76	1916	562	5	3.00	0.28	0.32	0.311
Komsomol'sk-on-Amur	50.53	136.98	–0.7	76	1859	499	10	8.10	0.89	0.27	0.286
Pobedino	49.83	142.81	–0.9	77	2221	604	~1	4.68	0.60	0.75	0.259
Yakovlevka	44.43	139.48	2.5	70	1877	626	~1	5.20	0.65	0.45	0.259
Pogranichnyi	44.40	131.15	3.6	67	1793	595	~1	6.28	0.61	0.31	0.256

Note: Corrosion rates expressed in *μm/year and **g/(m^2 year).

Table 13. Test sites (geographic coordinates), climate parameters, sulfur dioxide concentrations, chloride deposition rates, and rates of corrosion of structural metals in coastal regions at Barents Sea, East-Siberian Sea, Chukchi Sea, Bering Sea, Okhotsk Sea, and Japan Sea

Test site	Geographic coordinates		T, °C	RH, %	TOW, h/year	Precipitation, mm/year	$[SO_2]$, μg/m^3	$[Cl^-]$, mg/(m$_2$ day)	Corrosion rate			
	Latitude	Longitude							Steel*	Zinc*	Copper*	Aluminum**
Murmansk	69.15	35.97	0.5	82	3007	735	6.4	35–65	29–34	1.42–2.19	1.50–2.70	6.6–10.6
Aion	69.83	168.67	–13.0	88	1197	179	~1	33.8	8.3	1.61	0.63	1.71
Apapel'khino	69.73	170.62	–12.7	83	1451	193	~1	20.1	4.8	0.85	0.44	0.65
Arman'	55.77	156.17	–3.9	69	2115	415	5.0	5.9	4.2	0.84	0.52	0.35
Okhotsk	59.33	143.25	–5.0	74	2276	555	15.0	20.0	11.7	1.18	1.21	0.89
Ust'-Khar'yuzovo	57.18	165.75	–1.9	85	3014	512	~1	16.3	11.8	1.11	0.86	0.69
Ayan	56.25	138.33	–3.3	68	2251	791	~1	12.7	6.8	0.95	0.56	0.41
Chumikhan	54.67	135.25	–3.9	77	2075	681	~1	17.9	7.3	0.91	0.86	0.79
Baidukov	53.32	141.47	–2.5	83	2432	524	~1	32.1	14.5	1.65	1.90	1.42
Okha	53.58	143.02	–2.4	83	2322	476	5.0	10.3	8.6	1.12	0.90	0.62
Nikol'skoe	55.23	166.07	2.0	85	3798	636	~1	34.2	28.4	2.63	2.12	2.31
Lopatka	50.93	156.67	1.0	90	4760	2502	5.0	45.2	31.7	3.20	2.80	2.61
Petropavlovsk-Kamchatskii	53.05	158.72	1.9	72	2386	1092	15.0	11.4	13.7	2.14	1.08	0.75
Nevel'sk	46.68	141.92	4.5	74	3198	849	15.0	17.3	17.9	2.82	1.44	1.76
Vladivostok	43.15	131.88	4.7	75	3951	570	22.5	20–43	22–43	1.23–2.82	1.35–1.74	0.70–1.26
Cape Gamov	43.00	131.00	4.2	73	3497	728	~1	22.0	21.9	–	1.55	0.99

Note: Corrosion rates expressed in *μm/year and **g/(m^2 year).

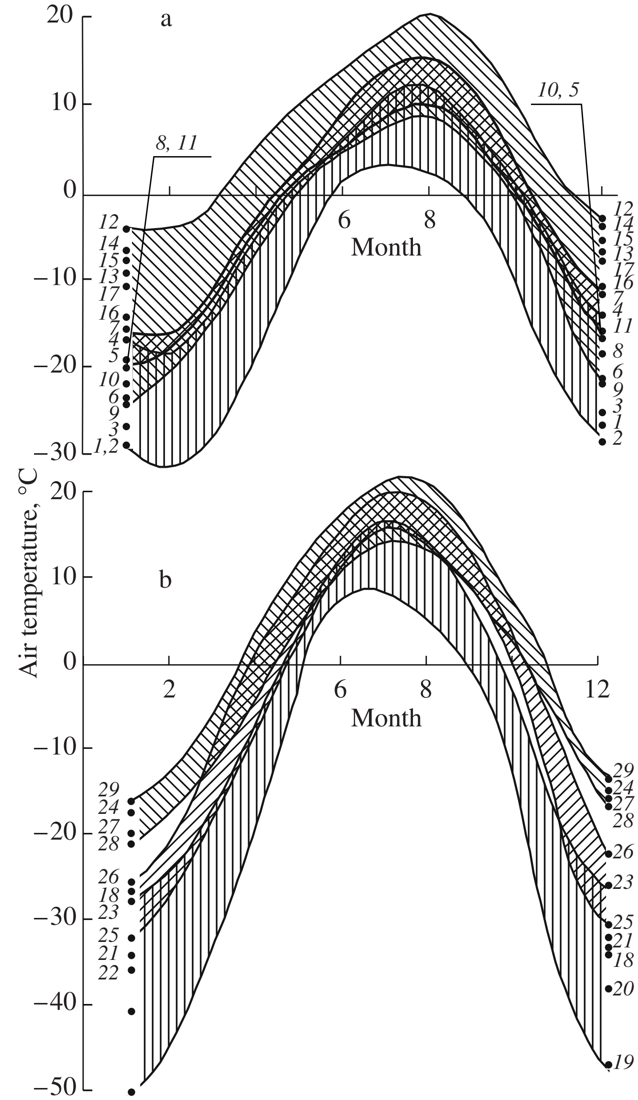

Fig. 6. The annual variation of the air temperature in various (a) coastal and (b) continental regions (numbers refer to stations indicated in Fig. 5 and in Tables 12 and 13): (*1–5*, *18–23*) Magadan oblast, Chukot National Territory; (*6–10*, *25*, *26*) Khabarovsk Krai, Amursk oblast; (*11–17*, *24*, *27–29*) Sakhalin Island, Kamchatka Peninsula, Primorskii Krai.

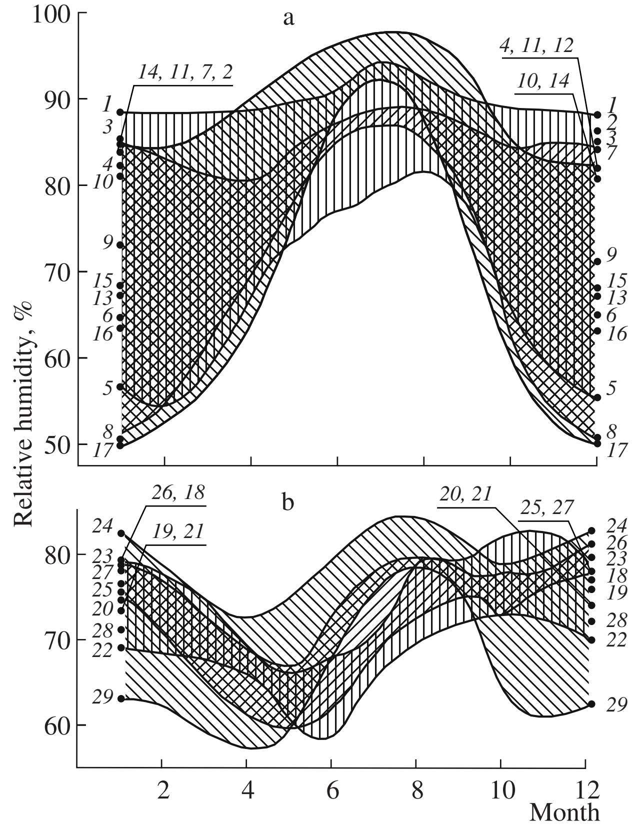

Fig. 7. Profiles of the annual variation of the relative humidity in various (a) coastal and (b) continental regions (notation is the same as in Fig. 6).

average annual temperature as low as –16.6°C) can be considered unique.

The rate of corrosion at temperatures from –16.6 to –11°C (Oimyakon, Susuman, Atka, Bilibino, Ust'-Omchug) for carbon steel (St3) was within 0.11–2.18 μm/year (to be compared with 0.05–0.87 μm/year for the continental stations of Antarctica). The intervals of corrosion rates for zinc and copper were 0.13–0.41 and 0.07–0.15 μm/year, respectively. In regions with somewhat warmer climate (south of Primorskii Krai, central regions of Kamchatka Peninsula and Sakhalin Island) and rural atmosphere, carbon steel corroded at a rate of 3.0–6.28 μm/year (Table 12). At Komsomol'sk-on-Amur, the maximum rate of corrosion was 8.10 μm/year for steel and 0.89 μm/year for zinc. For copper, the maximum rate of corrosion was 0.75 μm/year (Pobedino). For aluminum, the spread of corrosion rates on the continental part of the Far East was rather insignificant.

In coastal regions of the Far East, where the annual average temperatures are relatively low, the corrosion rates of carbon steel were approximately the same as in the coastal regions of Antarctica, while the corrosion rates of copper were even lower than those observed at the continental Antarctic station of Mirnyi. The maximum corrosion rates for coastal stations were observed at Vladivostok (Fig. 5, point *16*), Nevel'sk (*15*), Petropavlovsk-Kamchatskii (*13*), Cape Shmidt (*3*), and Baidukov Island (*10*). On the coastal stations at Apapel'khino Peninsula (*2*), Arman' (*5*), and Ayan (*8*), where the climate is less wet and the distance to the coast line is about 1 km, the corrosion rates are much lower.

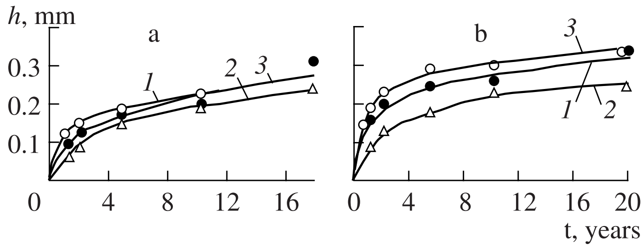

Fig. 8. Dependence of the maximum pit depth h in various aluminum alloys on the duration of exposure τ in (a) industrial atmosphere (Moscow) and (b) marine atmosphere (Barents Sea): (*1*) AMTsM, sheet; (*2*) AMg2P, sheet; (*3*) AD31T, profile [48].

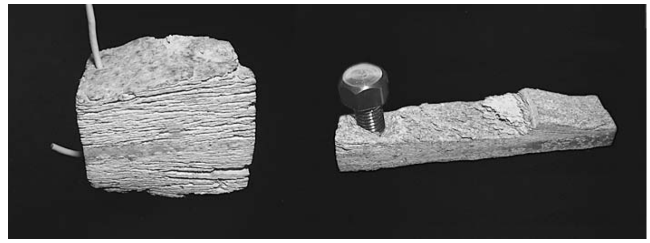

Fig. 9. Exfoliation corrosion and cracking of D16T alloy semiproducts tested for 14 years (1992–2006) at the NCS (Barents Sea coast).

Carbon steel was most sensitive to variations in the climate parameters: in continental regions, the corrosion rate of steel varied by a factor of 30, while that of zinc changed only by a factor of 7. Summarizing the data for steel, it can be concluded that the rates of corrosion in continental regions with clean (rural) atmosphere increases in cold macroclimate zones in the following order:

Antarctica < Far East < sub-Arctica.

This pattern is related primarily to differences in the temperature (and, hence, in the TOW of the metal surface) in the indicated cold territories. In comparison to regions with temperate climate (Europe), the rates of metal corrosion in the continental regions of Far East are much lower.

For aluminum, the character of corrosion development both in depth and on the surface significantly depend on the structure of initial blanks. Orientation of the metal structure along the main vector of straining favors preferential development of pitting in the surface layer.

SPECIFIC TYPES OF CORROSION

From practical standpoint, it is important not only to evaluate the general corrosion behavior of metals and alloys, but to assess the specific forms of corrosion as well. It was found that cold climate poses serious problems in this respect in both sub-Antarctic and sub-Arctic regions with marine atmospheres. For example, aluminum alloys used in constructions of the Bellingshausen station in Antarctica exhibited pronounced local corrosion. The laws of specific corrosion types have been studied for many years at the Northern Corrosion Station (NCS) of the Frumkin Institute of Physical Chemistry and Electrochemistry. In aluminum alloy D16T (artificially aged, not anodized), the average pit depth after a three-year exposure at this station in a marine atmosphere near Barents Sea was 26 μm, in comparison to 39 μm in the industrial atmosphere of Moscow and 17 μm in the rural atmosphere of Moscow region (Zvenigorod station) [47].

Figure 8 shows the dependence of the pit depth in aluminum alloys on the duration of exposure at the NCS [48]. In a box with baffle walls (modeling ventilated unheated room), the resistance of aluminum alloys to pitting corrosion was generally lower as compared to that in the industrial atmosphere: the maximum pit depth was greater by a factor of 2 for alloy 195T and by a factor of 2.3 for alloy AMg6M, while being approximately the same for D16T. A decrease in the corrosion rate was observed for loaded elements of flying vehicles made of high-strength aluminum alloys of the types D16, D1, D19, V95, AK6, AK8, etc. However, the exfoliation corrosion does not slow down with the time and, hence, is dangerous for flying vehicles, especially upon long (20–30 years) service [49, 50]. Starting at the surface, corrosion of this type proceeds in depth and leads to swelling and stratification of the metal. The corrosion products rapidly increase in volume and produce cleavage in the direction perpendicular to the direction of stratification (Fig. 9).

The process of crevice corrosion has a more local character as compared to that in an open atmosphere. We have performed atmospheric tests for crevice corrosion in carbon steel at the NCS. The crevice was formed by two metal plates rolled together at one end, making an angle of about 3°, and covered by a lacquer. After a one-year exposure, the lacquer coating in the crevice exhibited separation and surface corrosion took place within 2–3 cm from the contact site. In additional to the general corrosion, there were numerous pits, the number of which increased in the narrower part of the crevice. These pits reached 30–50 μm in lateral directions and 50–70 μm in depth.

Exfoliation corrosion is among the most frequently encountered types of corrosive damage in flying vehicles. Table 14 gives the coefficients of exfoliation corrosion rate enhancement (relative to that in rural atmosphere) for three high-strength aluminum alloys, which were determined using data on the depth of corrosion obtained in atmospheric tests conducted at various corrosion stations of the Frumkin Institute of Physical Chemistry and Electrochemistry. As can be seen, the maximum

Table 14. Coefficients of exfoliation corrosion enhancement in high-strength aluminum alloys [51, 52]

Test site	$[Cl^-]$, mg/(m^2 day)	Enhancement coefficient		
		D16	V95	AK6
Zvenigorod	0.1–0.3	1.0	1.0	1.0
Moscow	0.5–0.7	1.5	2.0	2.5
Batumi	2–3	1.5	2.0	17.0
Dal'nie Zelentsy	20–30	17.5	24.0	22.0

corrosion enhancement coefficients are observed at the NCS, where the marine atmosphere is characterized by the maximum concentration of chlorides.

In alloys such as V92T, V95T1, ATsMT, T1 and other high-strength alloys of the Al–Zn–Mg system, in contrast to those containing copper (D16, V95, AK6, etc.), cracking corrosion typically starts developing at the edge of a sample, which is deformed as a result of cutting sheet and thin profiles. This damage is manifested as separate cracks (resembling layer degradation of metals in the absence of external stresses). In marine atmospheres, the cracks develop (like the exfoliation corrosion) without any retardation and reach considerable depths (Fig. 10). However, in the absence of halides, the crack ceases to propagate after a two-year exposure, the maximum penetration depth reaching 3–4 mm.

Corrosion cracking is among the most dangerous forms of corrosive damage. Illustrative results showing the influence of environmental conditions on the crack propagation were obtained in tests on D16T (Fig. 9), V95T1 and AK4-1T [51]. The length of cracks tends to increase with increasing chloride content in the atmosphere (Fig. 11). Alloys of the D16, AK6, and AK4-1 proved to be most susceptible to the detrimental action of chlorides. In the absence of halide ions, the crack in D16 type alloys cease to grow even in industrial atmospheres, although strong pitting and/or integranular corrosion can proceed. In contrast to D16T, alloy V95pchT1 exhibits significant cracking even under conditions of a heated room ($RH < 70\%$). The effect of chlorides on the crack propagation is consistent with their influence on the exfoliation corrosion in high-strength aluminum-based alloys.

Thus, specific types of corrosion are most dangerous under cold climate conditions. The rates of such corrosion in cold climate zones, although generally lower than in tropical marine atmospheres, are still rather high and determined primarily by the presence of chlorides and the rate of their deposition.

GLOBAL WARMING AND CORROSION

In recent years, it is frequently predicted that a significant global warming will take place in the 21st century. There various opinions concerning true causes of this phenomenon, but the uncontrolled influence of human activity on the environment, including the ozone layer, seems to be among real factors. According to prognosis, the interval of annual average temperatures can shift up by 2–6°C. Accordingly, various scenarios of climate change in various regions of the Earth are

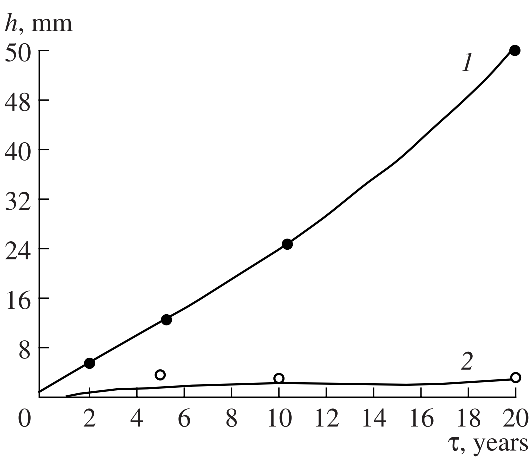

Fig. 10. Plots of the crack penetration depth h versus duration of exposure τ for ATsMT1 alloy (4.0-mm-thick profile) tested in (*1*) marine atmosphere (Barents Sea) and (*2*) industrial atmosphere (Moscow) [51].

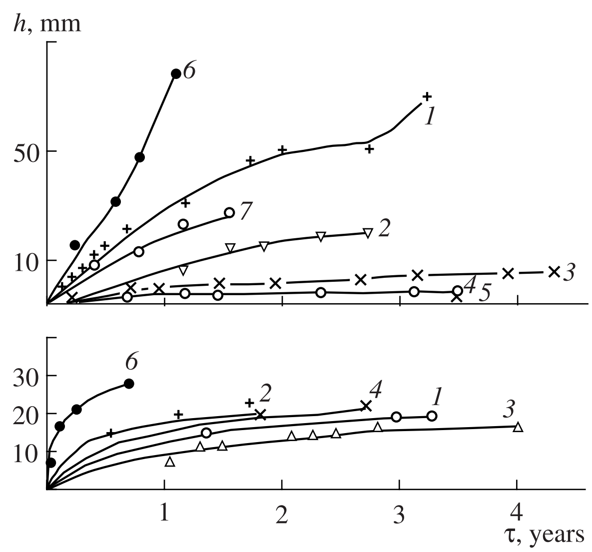

Fig. 11. Dependence of the crack length l in corrosion-cracked (a) D16chT alloy (plate) and 1915T alloy (pressed stripe) on the test duration at various sites: (*1*) Barents Sea coast (NCS); (*2*) Black Sea coast; (*3*) industrial atmosphere; (*4*) heated room; (*5*) rural atmosphere; (*6*) tropical marine atmosphere; (*7*) Okhotsk Sea coat [51].

outlined and possible consequences of this global warming are predicted.

The authors of various scenarios have a common point, according to which the warming will also involve cold regions and cause the melting of glaciers in both continental regions and seas.

From the standpoint of corrosion, this will favor an increase in the real TOW of metal surfaces. Accordingly, it is highly probable that the rate of corrosion in presently cold and very cold regions of the Earth will increase.

Calculations performed using models developed at the Frumkin Institute of Physical Chemistry and Electrochemistry within the framework of the Program "New Approaches to Corrosion and Radiation Resistance of Materials and Radioecological Safety" (Department of Chemical Sciences, Russian Academy of Sciences) showed that an increase in the annual average temperature by 3°C will lead (under otherwise equal conditions) to an increase in the rate of corrosion, in particular, of carbon steel (by 25%), weathering steel (19%), copper (28%), bronze (20), and zinc (4.5%). If this increase in the annual average temperature will be accompanied by a 5% increase in the relative humidity, the acceleration of corrosion will be even more pronounced, reaching 61% for steel (St3), 32% for weathering steel, 36% for copper, 27% for bronze, and 16% for zinc.

According to models developed within the framework of the United Nations programs, a 3°C increase in the annual average temperature will increase the corrosion loss of materials in electric contacts by 67 and 39% for nickel and tin, respectively.

The above examples show that an increase in the rate of corrosion caused by a global warming of about 3°C is not catastrophic, but yet significant, in particular, for electric contact materials (nickel, tin) and carbon steels.

CONCLUSIONS

We have considered the results of atmospheric corrosion tests on various metals and alloys in the Earth's regions with cold and very cold climate (Antarctic, sub-Arctic, Russian Far East). These data and their analysis lead to the following important conclusions.

1. The corrosion of metals is observed even in very cold climate with negative annual average temperatures, including sites such as the Amundsen–Scott south-pole station and Vostok (pole of cold) station in Antarctica and continental stations on the Far East territories (Oimyakon, Bilibino, etc.). This is evidence, in particular, for the absence of water freezing in the adsorbed phase even at low temperatures—in agreement with the results of laboratory investigations, which show that adsorbed water does not exhibit freezing in the region of mono- and polymolecular adsorption.

2. Investigations in sub-Arctic regions such as Sulitjelma (Northern Norway) and near the Russia–Norway border gave close results concerning the corrosion of carbon steel and zinc. According to this, the lower temperature threshold for the development of corrosion in these regions is at –4°C.

3. Carbon steel is most sensitive among all metals studied with respect to environmental characteristics in the regions of cold and very cold. In continental regions of the Far East, the corrosion rate of carbon steel exhibits 30-fold variations, while the corrosion rate of zinc varied only by a factor of 7.

4. Experimental data on the corrosion of steel in continental regions with clean (rural) atmosphere increases in cold macroclimate zones in the following order:

Antarctica < Far East < sub-Arctica.

This is related primarily to differences in the temperature (and, hence, in the TOW of the metal surface) in the indicated cold territories.

5. The contamination of atmosphere with sulfur dioxide can lead, even in cold sub-Arctic climate, to a significant increase in the rate of metal corrosion. In terms of the international standard (ISO 9223), the corrosivity of such contaminated atmospheres can be rated C3 and even C4 (near the source of SO_2 emission).

6. The transfer and deposition of chlorides in the marine atmospheres of sub-Antarctic, sub-Arctic, and Russian Far East can lead to a significant acceleration of metal corrosion. Most dangerous are some specific types of corrosion, including pitting, exfoliation corrosion, crevice corrosion, and corrosion-induced cracking. The intensity of specific corrosion types is determined primarily by the rate of chloride deposition and remains rather high even in the regions of cold climate with low average air temperatures.

7. Approximate estimates of the influence of global warming on the atmospheric corrosion of metals are obtained. In the regions of cold climate. an increase in the rate of corrosion caused by a global warming of about 3°C is not catastrophic, but yet significant, in particular, for electric contact materials (nickel, tin) and carbon steels. Indeed, under conditions modeling the device operation in a nonheated ventilated room, the rate of corrosion increases by 67% for nickel and by 39% for tin. The situation will be more complicated if an increase in the temperature due to global warming will be accompanied by an increase in the relative humidity of air.

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